Software Development Project — G&G City · Lab Backlog Sprint 1

LAB C2: Cat Skills + Speech Bubble — Adnan Masri

Topic: Chaining Animations with Promises + a Bubble that Follows Its Owner

Learning outcomes

By the end of this lab, students will be able to:

- Wrap a CSS animation in a `Promise` so others can write `await cat.charge()`
- Restart the same CSS animation more than once
- Position an element relative to a moving parent with `position: absolute`

🎯 Problem Scenario

The cat can stand and walk (C1). Now it needs magic: every time the learner runs a command, the cat charges, then casts a green or red spell, then talks.

- **Start after C1 pushes `createCat`, then `git pull`**
- **Write only inside `function addSkills(cat)` at the end of `cat.js` (C1 already calls it)**
- **S3 will call: `await cat.charge(); await cat.spell(ok); cat.say(text);`**

Your files: `cat/cat.js (C2 section only), cat/test.html`

Required Outputs

Build:

1. **`charge()`: arm goes up + golden glow (≈0.9s), returns a Promise**
2. **`spell(ok)`: green flash when true, red flash + shake when false (≈0.7s), returns a Promise**
3. **`say(text, ms)`: bubble above the head that follows the cat, supports Arabic, `ms = 0` keeps it visible**
4. **Add buttons for the three skills to `test.html`**

Code is given only for the hard parts. Parts marked 📝 **Note** are easy: follow the note and write them yourself. Test with **Live Server** in VS Code (`fetch` does not work when you double-click a file).

Model Answer:

Task 1 + 2: The animations (CSS)

> 📝 **Note (no code needed):** Write a SKILL_CSS string and inject it once, exactly like C1 did with CAT_CSS. You need three classes on `cat.el`: `.is-charging` (rotate `#cat-arm` and grow a yellow `drop-shadow`), `.spell-ok` (green `drop-shadow` flash) and `.spell-fail` (red flash + `translateX` shake). Give `#cat-arm` a `transform-origin` at the shoulder.

Task 1 + 2: Turning a CSS class into a Promise (hard part)

Removing the class, forcing a reflow with `offsetWidth`, and adding it again is what lets the same spell play twice in a row.

```
// Plays a CSS class once and resolves when it is done → lets S3 write: await cat.charge()
function play(cls, ms) {
  return new Promise(resolve => {
    cat.el.classList.remove(cls);
    void cat.el.offsetWidth;                          // restart the animation if it is
replayed
    cat.el.classList.add(cls);
    setTimeout(() => { cat.el.classList.remove(cls); resolve(cat); }, ms);
  });
}

cat.charge = () => play('is-charging', 900);
cat.spell  = ok => play(ok ? 'spell-ok' : 'spell-fail', 700);
```

Task 3: The speech bubble (hard part)

The bubble is a child of `cat.el`, so when S3 moves the cat, the bubble moves with it for free. `dir="auto"` picks right-to-left for Arabic automatically.

```
// The bubble lives INSIDE cat.el, so it follows the cat wherever S3 moves it
const bubble = document.createElement('div');
bubble.className = 'gg-bubble';
bubble.dir = 'auto';                                  // Arabic → RTL, English → LTR
automatically
cat.el.appendChild(bubble);

let hideTimer;
cat.say = (text, ms = 4000) => {                      // ms = 0 → stays until the next
say()
  bubble.textContent = text;
  bubble.classList.add('show');
  clearTimeout(hideTimer);
  if (ms) hideTimer = setTimeout(() => bubble.classList.remove('show'), ms);
```

```
  return cat;
};
```

> 📝 **Note (no code needed):** Style `.gg-bubble` with `position: absolute; bottom: 100%; left: 50%`, a border, rounded corners, `opacity: 0` by default and `opacity: 1` on `.show`. Add `white-space: pre-line` so `\n` makes a new line. The little tail is a `::after` with a border triangle. Set `.gg-cat { position: relative }`.

Task 4: test.html

> 📝 **Note (no code needed):** Add buttons: Charge, Spell ✓, Spell ✗, and a text input + button that calls `cat.say(input.value)`. Try clicking Spell ✗ many times fast: it must replay every time.

Expected Output



charge() → gold glow



spell(false) → red



spell(true) → green

Submission

```
git pull
git add cat/
git commit -m "إضافة شحن الطاقة والتعويذة وفقاعة الكلام"
git push
```